



# ð'ð³/₄ð±ð° / Boba

Multi-tracker meta-search for qBittorrent â€” self-hosted, containerised, private-tracker-aware.

[Quick start](#) Â· [Tokens & keys](#) Â· [Documentation](#) Â· [Architecture](#) Â· [Testing](#) Â· [Contributing](#)

python tests 585 passing frontend tests 182 passing plugins 48 merge service FastAPI :7187  
ci auto (syntax / unit / integration / nightly / security)  
scanners snyk | sonar | bandit | ruff | semgrep | trivy | gitleaks | pip-audit license Apache 2.0

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## Features

- **Merge Search Service** â€” FastAPI service (:7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
- **Real-time results** â€” result\_found events arrive as each tracker completes, no blocking.
- **Private-tracker bridge** â€” Authenticated downloads via webui-bridge.py for RuTracker, Kinozal, NNM-Club, IPTorrents.
- **Freeleech-only IPTorrents** â€” Automation never costs ratio (see [constitution VIII](#)).
- **Opt-in quality stack** â€” SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
- **Opt-in observability** â€” Prometheus + Grafana dashboards behind the same profile system.
- **Hook system** â€” Register scripts via POST /api/v1/hooks for search / download events.
- **Angular 21 SPA dashboard** â€” dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
- **PWA-ready** â€” favicon + launcher icons + web manifest ship from frontend/public/.

- **Container-first** – rootless Podman or Docker, single `./start.sh` boot.

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## Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent

# Optional: configure tracker credentials + tokens
cp .env.example .env
$EDITOR .env                # see "Tokens & API keys" below

./setup.sh                  # one-time
./start.sh                  # Angular build + containers + plugin sync
```

Then visit:

| URL                                                                                 | Purpose                                  | Auth          |
|-------------------------------------------------------------------------------------|------------------------------------------|---------------|
| <a href="http://localhost:7187/">http://localhost:7187/</a>                         | Merge-search dashboard (SPA)             | –             |
| <a href="http://localhost:7187/docs">http://localhost:7187/docs</a>                 | FastAPI Swagger UI                       | –             |
| <a href="http://localhost:7187/openapi.json">http://localhost:7187/openapi.json</a> | OpenAPI schema                           | –             |
| <a href="http://localhost:7186/">http://localhost:7186/</a>                         | qBittorrent WebUI proxy                  | admin / admin |
| <a href="http://localhost:7185/">http://localhost:7185/</a>                         | qBittorrent internal (container)         | –             |
| <a href="http://localhost:7188/">http://localhost:7188/</a>                         | WebUI bridge (private-tracker downloads) | –             |

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## Tokens & API keys

– **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS\\_AND\\_KEYS.md](#) – which variable is mandatory, which is optional, and **where to register** for each one.

### At a glance

| Category              | Mandatory   | Optional                       | Documentation      |
|-----------------------|-------------|--------------------------------|--------------------|
| qBittorrent WebUI     | –           | WEBUI_USERNAME, WEBUI_PASSWORD | <a href="#">§1</a> |
| Private trackers      | per-tracker | –                              | <a href="#">§2</a> |
| Public tracker uplift | –           | JACKETT_API_KEY, –             | <a href="#">§3</a> |

| Category            | Mandatory | Optional                                      | Documentation       |
|---------------------|-----------|-----------------------------------------------|---------------------|
| Metadata enrichment | â€”       | TMDB_API_KEY, TVDB_API_KEY, â€”               | <a href="#">Â§4</a> |
| Security scanning   | â€”       | SNYK_TOKEN, SONAR_TOKEN, â€”                  | <a href="#">Â§5</a> |
| Observability       | â€”       | GRAFANA_USER, GRAFANA_PASSWORD                | <a href="#">Â§6</a> |
| Orchestrator tuning | â€”       | ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, â€” | <a href="#">Â§7</a> |

## Where to register (fast links)

| Provider    | Signup                                                                                                      | Notes                           |
|-------------|-------------------------------------------------------------------------------------------------------------|---------------------------------|
| RuTracker   | <a href="https://rutracker.org/forum/register.php">https://rutracker.org/forum/register.php</a>             | Username + password             |
| Kinozal     | <a href="https://kinozal.tv/signup.php">https://kinozal.tv/signup.php</a>                                   | May require invite              |
| NNM-Club    | <a href="https://nnmclub.to/forum/ucp.php?mode=register">https://nnmclub.to/forum/ucp.php?mode=register</a> | Cookie-based auth               |
| IPTorrents  | <a href="https://iptorrents.com/">https://iptorrents.com/</a>                                               | Invite-only; freeleech enforced |
| Jackett     | <a href="https://github.com/Jackett/Jackett">https://github.com/Jackett/Jackett</a>                         | Self-hosted indexer             |
| TMDb        | <a href="https://www.themoviedb.org/signup">https://www.themoviedb.org/signup</a>                           | v3 auth API key                 |
| TVDb        | <a href="https://thetvdb.com/api-information">https://thetvdb.com/api-information</a>                       | Subscription API                |
| MusicBrainz | <a href="https://musicbrainz.org/doc/MusicBrainz_API">https://musicbrainz.org/doc/MusicBrainz_API</a>       | Free â€” needs UA only          |
| Snyk        | <a href="https://app.snyk.io/">https://app.snyk.io/</a>                                                     | Account â†’ Auth Token          |
| SonarCloud  | <a href="https://sonarcloud.io/account/security/">https://sonarcloud.io/account/security/</a>               | Or run the compose SonarQube    |
| Gitleaks    | <a href="https://gitleaks.io/">https://gitleaks.io/</a>                                                     | Optional commercial license     |

## Documentation

Everything the platform offers, indexed:

### Tracked-Items + Status Documents

The mandated entry point (Â§11.4.57 / Â§11.4.212) for every tracker and status document in this repository â€” Last modified and Revision are

sourced verbatim from each document’s own revision header (Â§11.4.44); a “ means the document currently carries no revision header or has no HTML/PDF export yet (never fabricated).

| Document                                | Last modified        | Revision | Markdown                 | HTML                 |
|-----------------------------------------|----------------------|----------|--------------------------|----------------------|
| Issues                                  | 2026-06-09T20:00:00Z | 5        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| Issues_Summary                          | “                    | “        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| Fixed                                   | 2026-06-09T21:00:00Z | 13       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| Fixed_Summary                           | “                    | “        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| CONTINUATION                            | 2026-06-16T23:55:00Z | 19       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| PORTING-FROM-LAVA                       | 2026-07-01T16:18:43Z | 1        | <a href="#">Markdown</a> | “ (n<br>yet<br>expor |
| REMAINING_WORK_PLAN                     | 2026-08-07T19:10:54Z | 2        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| GOVERNANCE_AUDIT_2026-08-07             | 2026-08-07T19:10:54Z | 1        | <a href="#">Markdown</a> | “ (n<br>yet<br>expor |
| browser_extension/Status                | 2026-06-13T13:10:00Z | 15       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| browser_extension/<br>Status_Summary    | 2026-06-10T20:05:00Z | 3        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| browser_extension/<br>RELEASE_READINESS | 2026-06-13T13:10:00Z | 4        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| COMPLETION_STATUS                       | 2026-06-06T00:00:00Z | 1        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| RELEASE_READINESS_20260616              | 2026-06-16T10:30:00Z | 1        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| features/Status                         | 2026-06-16T23:30:00Z | 7        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| features/Status_Summary                 | 2026-06-16T23:30:00Z | 7        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| QA_DISCOVERY_LEDGER                     | 2026-08-09T12:45:00Z | 2        | <a href="#">Markdown</a> | <a href="#">HTML</a> |

## Getting started & operation

- [docs/USER\\_MANUAL.md](#) “ end-user walkthrough
- [docs/TOKENS\\_AND\\_KEYS.md](#) “ “ **credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN\\_SYSTEM.md](#) “ runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) “ the 48 plugin engines
- [docs/PLUGIN\\_TROUBLESHOOTING.md](#) “ what to check when a plugin breaks

## Architecture & subsystems

- [docs/architecture/](#) “ 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA\\_MODEL.md](#) “ Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) “ asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) “ Prometheus + Grafana + OpenTelemetry

- [docs/PERFORMANCE.md](#) â€” benchmark + load + stress test layout
- [docs/api/openapi.json](#) â€” frozen OpenAPI spec (diffed in CI)

## Quality & security

- [docs/SECURITY.md](#) â€” threat model + credential storage
- [docs/SCANNING.md](#) â€” Snyk + Sonar + Semgrep + Trivy + Gitleaks + bandit + pip-audit
- [docs/QUALITY\\_STACK.md](#) â€” the opt-in docker-compose.quality.yml stack

## Testing

- [docs/TESTING.md](#) â€” catalogue of every test type (30 rows)
- [docs/COVERAGE\\_BASELINE.md](#) â€” per-module coverage gates
- [tests/README.md](#) â€” layout of the tests/ tree

## Development

- [CONTRIBUTING.md](#) â€” contribution workflow
- [CLAUDE.md](#) â€” Claude Code agent protocol (TDD + rebuild-reboot)
- [AGENTS.md](#) â€” runtime development guidance
- [.specify/memory/constitution.md](#) â€” **binding architectural contract (v1.1.0)**

## Courses (self-paced, Asciiinema)

- [courses/01-operator/](#) â€” Your first search
- [courses/02-plugin-author/](#) â€” Authoring a nova3 plugin
- [courses/03-contributor/](#) â€” TDD + rebuild-reboot deep dive
- [courses/04-security-ops/](#) â€” Threat model + scanner bundle

**Release / CI**

- [CHANGELOG.md](#)
- [releases/README.md](#) â€” release artefacts layout
- [docs/OUT\\_OF\\_SANDBOX.md](#) â€” items requiring external credentials (HelixQA / OpenCode / submodule orgs)

# Architecture

[illegible]

FastAPI

### Opt-in (docker-compose.quality.yml):

Full Mermaid renders in [docs/architecture/](#).

```
# Non-live-HTTP suites " run anywhere, ~10s
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q

# Benchmarks " dedupe perf + live-HTTP fan-out benchmarks, ~4min
python3 -m pytest tests/benchmark/ --no-cov -q

# Security " hits the live merge service, needs stack up
python3 -m pytest tests/security/ --no-cov -q

# Integration " same
python3 -m pytest tests/integration/ --no-cov -q -m "not requires_credentials"

# Frontend
npx --prefix frontend ng test --watch=false

# Full scanner sweep (non-interactive; skips scanners with missing tokens)
./scripts/scan.sh --all
```

## Releases

```
./scripts/build-releases.sh          # all targets, all channels
./scripts/build-releases.sh frontend # one target
./scripts/build-releases.sh --channel release
```

# Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (pytest â€” 585+ tests)
  - Frontend Vitest (ng test â€” 182+ tests)
  - Ruff + bandit + shellcheck (via `scripts/scan.sh`)
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## License

Apache 2.0 â€” see [LICENSE](#).